

Test and Evaluation Department

Ballistic Testing Facility

Test Report

Ballistic Resistance Test

Bullet Proof Sliding Gate (Main Gate)

Drawing No. "SSW-002 Dated 17 July 2022"

EN1522/1523 Standard Level (FB6)

7.62x51mm NATO Ball "DM111" & 5.56x45mm "SS109"

Special And Safety Work Nisreen Factory / KSA

JODDB/TEST/BTF/TR/1089



4515

This report relates only to the test specimens tested and should not be published or copied or distributed or reissued to any party whatsoever without the prior written permission of Jordan Design and Development Bureau

Approval

Test Report

Ballistic Resistance Test

Bullet Proof Sliding Gate (Main Gate)

Drawing No. "SSW-002 Dated 17 July 2022"

EN1522/1523 Standard Level (FB6)

7.62x51mm NATO Ball "DM111" & 5.56x45mm "SS109"

Special And Safety Work Nisreen Factory / KSA

Compiled by

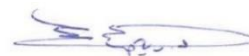
Head of Weapons, Ammunition & Armor Testing

Eng. Issa Abed AL Rahman Rawashdeh

Approved by

Head of Test & Evaluation Department

Dr. Eng. Riyadh Ali Ratrouf



Signature

20 September 2022

Date

1. Test Data

Requester	JODDB / Sales
Contractor	Special And Safety Work Nisreen Factory HRMM+WHV, As Samh Ibn Malik, Ad Difa, Riyadh 14315, KSA
References	- Test and Evaluation Request No. 1089 dated 31/08/2022 - Test Proposal No .JODDB/TEST/BTF/TP/1089 Dated 31/08/2022 (See Appendix A)
Test Sample	Bullet Proof Sliding Gate (Main Gate) Drawing No. SSW-002 Dated 17 July 2022 (See Photo No. 1)
Manufacturer	Special And Safety Work Nisreen Factory / KSA
Test Type	Ballistic Resistance Test
Test Standard	Windows, doors, shutters and blinds - Bullet resistance Requirements and classification EN 1522-1523 FB6
Test Site	Ballistic Testing Facility (JODDB)
Samples Received Date	21 th August 2022
Test Temperature	23 °C
Test Weapon	- NATO Universal Ballistic Breech with standard barrel: 7.62x51 mm ,SN:5043 - NATO Universal Ballistic Breech with standard barrel: 5.56x45 mm ,SN:3720
Type of Bullets	7.62x51mm NATO Ball “DM111” full metal jacket (steel), Pointed Bullet, lead-soft core Bullet Mass (9.55 ± 0.1) gram. Batch. No. 140423 5.56x45mm “SS 109” Full Steel Jacket, Pointed Bullet lead-soft core steel penetrator, Bullet Mass (4.0 ± 0.1). Batch. No. 605-FNB-08
Test Equipment	- Doppler Radar SN.3503-13-270-011 (± 0.4 m/s) - Digital caliper SN.A18229682 (± 0.05 mm) - Temp .Hum. Meter SN.95941255 (± 0.3 °C) - Distance laser Measuring Device (Lacia) SN. 1030969032 (±2 mm)
Test Conducted By	- Eng. Ali Sardyah - Eng. Sameer Obeidat - Eng., Mohannad Akilan - Senior Technician: Jalal Ghulilat - Technician : Tareq Almomani - Technician: Islam Alqurashi
Test Date	11 th September 2022

Note: The results contained in this report are only valid for detailed above and the report shall not be reproduced except full without approval of the laboratory.

2. Test Results

Test Sample Bullet Proof Sliding Gate (Main Gate) Drawing No. SSW-002 Dated 17 July 2022 EN 1522/1523 "FB6" (7.62x51mm NATO Ball "DM111") & (5.56x45mm "SS109")				
Threat	Shot No.	Bullet Velocity @ 7.5m (m/s)	Angle of Incidence (°)	Status(P/NS/S) P- Perforation NS-No perforation & No Splinters S - No Perforation with Splinters
7.62x51mm NATO Ball "DM111" Bullet Velocity (830 ±10)m/s @ 7.5m	1	829.3	0	NS
	2	839.1	0	NS
	3	835.9	0	NS
	4	825.5	0	NS
	5	835.3	0	NS
	6	827.3	0	NS
	7	827.2	0	NS
	8	823.9	0	NS
	9	830.6	0	NS
	10	830.6	0	NS
	11	828.9	0	NS
	12	835.1	0	NS
	13	843.5*	0	NS
5.56x45mm "SS 109" Bullet Velocity (950 ±10)m/s @ 7.5m	1	941.5	0	NS
	2	944.1	0	NS
	3	946.0	0	NS
	4	940.3	0	NS
	5	941.2	0	NS
	6	952.8	0	NS
	7	952.3	0	NS
	8	944.7	0	NS
	9	945.6	0	NS
	10	949.3	0	NS
	11	948.5	0	NS
NS-No perforation & No Splinters. See photos No.4				
Status: Comply				
EN1522/1523 Performance Requirements P - Perforation: Piercing of a test piece by a bullet or by bullet fragments, and/or creation of an opening from the attack face to the rear face The following are considered to constitute perforation: a) passing through the test specimen by the bullet or any part of it; b) splitting of the rear surface of the test specimen by the bullet or part of it, even if the bullet is visibly retained in the rear of the test specimen; c) Creation of an opening right through the specimen, even if that opening closes again afterwards. . S - No Perforation with Splinters: No perforation of the test piece by the bullet or parts of the bullet. But with perforation of the witness foil by glass splinters from the protected face. NS – No Perforation & No Splinters: No perforation of the test piece by the bullet or parts of the bullet. And no perforation of the witness foil by glass splinters from rear face.				

* Velocity is above the limit but considered fair hit since No penetration is occurred

RANGE EQUIPMENT & CONFIGURATION

THE GUN

The test sample were shot from the NATO Universal Ballistic Breech with the appropriate standard barrel and bullet type to give projectile stability.

VELOCITY MEASUREMENT

The projectile velocity was measured using Doppler radar system at 7.5 m from the muzzle with uncertainty 0.4 m/s.

See Applied Statement of Conformity and Decision Rule

Test Level	Test Result	Applied Decision Rule	Statement of Conformity
EN1522/1523 (FB6) 7.62x51mm NATO Ball "DM111" 5.56x45mm "SS109"	No Penetration	No perforation of the test piece by the bullet or parts of the bullet and No perforation of the witness foil When "DM111" Bullet velocity@7.5m ≥ 821.4 m/s. "SS 109" Bullet velocity @7.5m ≥ 941.4 m/s	Comply

DISTANCE MEASUREMENT:

Test range distance was measured by laser measuring device with uncertainty ± 2 mm, striking distance were marked by a digital caliber with uncertainty ± 0.05 mm.

SUPPORT FIXTURE:

The test sample were supported by a fixture that permits its position and attitude to be readily adjust so that it is perpendicular to the line of the flight of the bullet at the point of impact.

AIMING:

The aiming process has been achieved by using standard laser bullet.

SHOT PLACEMENT:

All shots hit the test sample in the required pattern, with the required place.

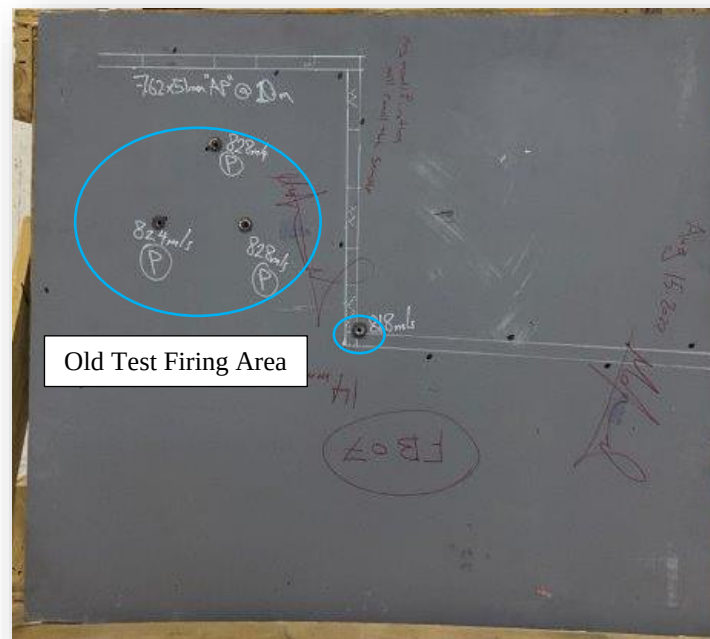
SPLINTER COLLECTING BOX & WITNESS FOIL:

It is fully enclose the space between the test piece and the 0.02mm thickness standard witness foil with density 54 g/m² which is mounted in the splinter collecting box in a vertical position at a distance of (500-+10)mm behind the test piece according to EN1522/1523 standards.

Photo No. 1: Test Sample (Before Firing)

Bullet Proof Sliding Gate (Main Gate) Drawing No. SSW-002 Dated 17 July 2022

Front view



Isometrec veiw



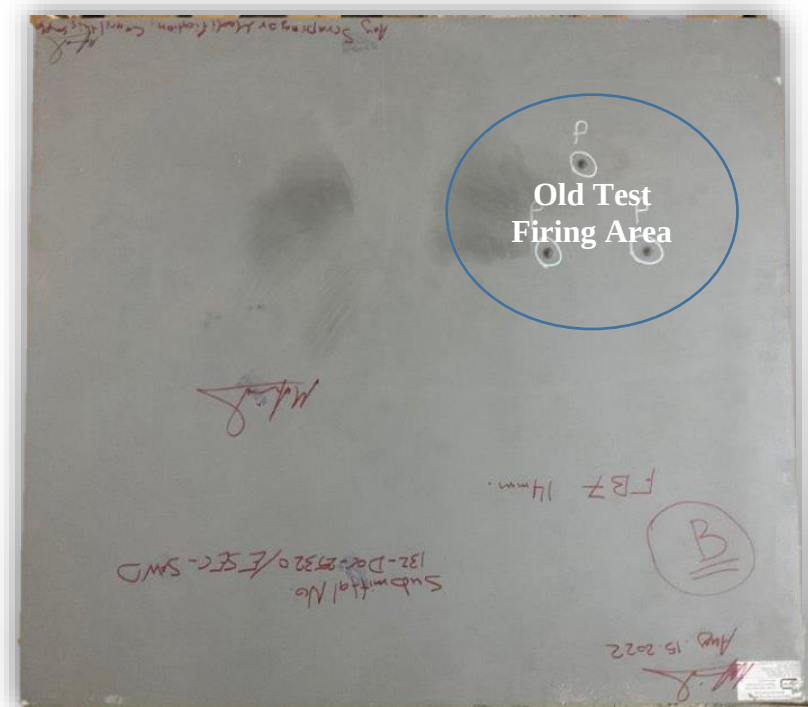
RESTRICTED

Photos No. 2: Status of Bullet Proof Sliding Gate (Main Gate) after firing 7.62x51mm NATO Ball “DM111” & 5.56x45mm “SS109” rounds from 10 m.

Bullet Proof Sliding Gate (Main Gate) (No Penetration)
Impact Side



Back Side



APPENDIX A: Approved Technical Testing Proposal

JODDB
المركز الأردني للتصميم والتطوير
JORDAN DESIGN & DEVELOPMENT BUREAU

المركز الأردني
للتصميم والتطوير

RESTRICTED

Test and Evaluation Department
Ballistic Testing Facility
Technical Testing Proposal

Contractor	Special And Safety Work Nisreen Factory / KSA			
Requester	JODDB/ Sales			
Reference	Test and Evaluation Request No. 1089 dated 31/08/2022			
Test Sample	Test Type	Test Reference	Test Requirements	Test Criteria
Bullet Proof Sliding Gate (Main Gate) QTY (1) Drawing No. SSW-002 dated 17 July 2022. (See Annex B)	Ballistic Resistance	Windows, doors, shutters and blinds - Bullet resistance Requirements and classification EN 1522-1523 FB6 "Accredited Test"	Firing rounds type 7.62x51mm NATO Ball "DM111" and 5.56x45mm "SS109" on impact side at each selected target area decided by testing lab as shown in Annex B. Test Range: 10±0.5 m Bullet Velocity - 7.62x51mm NATO Ball "DM111" @7.5m (Doppler): 830±10 m/s Bullet Velocity - 5.56x45mm "SS109" @7.5m (Doppler): 950±10 m/s	See Annex A

Test & Evaluation Department
JODDB/TEST/BTF/TP/1089
ص.ب 928125، عمان 11190، الأردن
Tel. : +962 (0) 6 4603230 - Fax. : +962 (0) 6 4603243 www.joddb.com

Page 1 of 3

August 31, 2022

JODDB
المركز الأردني للتصميم والتطوير
JORDAN DESIGN & DEVELOPMENT BUREAU

المركز الأردني
للتصميم والتطوير

RESTRICTED

Annex A

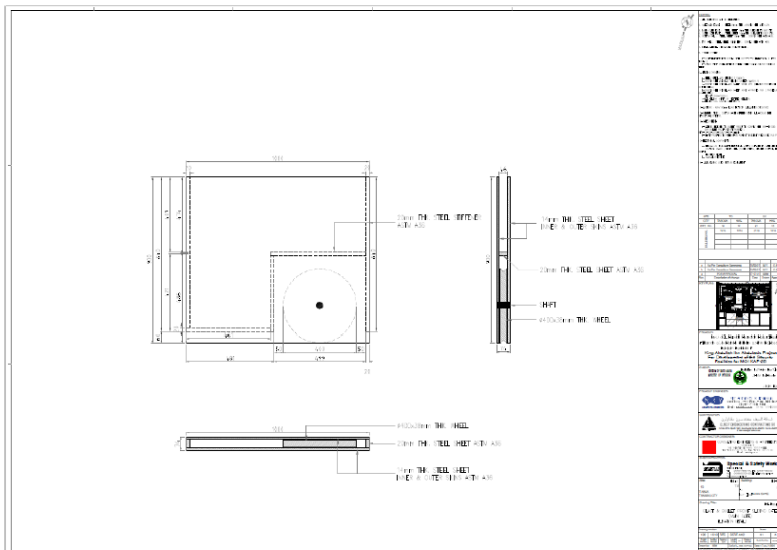
Test Criteria Statement of Conformity and Decision Rule			
Test Level	Comply	Comply with Conditions	Not Comply
EN1522/1523 FB6 7.62x51mm "DM111" 5.56x45mm "SS109"	No perforation of the test piece by the bullet or parts of the bullet and No perforation of the witness foil when "DM111" Bullet velocity@7.5m ≥ 821.4 m/s. "SS 109" Bullet velocity @7.5m ≥ 941.4 m/s		Perforation of the test piece by the bullet or parts of the bullet or Perforation of the witness foil when "DM111" Bullet velocity@7.5m ≤ 838.60 m/s. "SS 109" Bullet velocity@7.5m ≤ 958.6 m/s Note: The following are considered to constitute perforation: a) Passing through the test specimen by the bullet or any part of it; b) Splitting of the rear surface of the test specimen by the bullet or part of it, even if the bullet is visibly retained in the rear of the test specimen; c) Creation of an opening right through the specimen, even if that opening closes again afterwards.

Test & Evaluation Department
JODDB/TEST/BTF/TP/1089
ص.ب 928125، عمان 11190، الأردن
Tel. : +962 (0) 6 4603230 - Fax. : +962 (0) 6 4603243 www.joddb.com

Page 2 of 3

August 31, 2022

Annex B.



Test & Evaluation Department
JODDB/TEST/BTF/TP/1089
P.O Box 928125, Amman 11190, Jordan ص.ب. 928125، عمان 11190، الأردن
Tel. : +962 (0) 6 4603230 - Fax. : +962 (0) 6 4603243 www.joddb.com

Page 3 of 3

August 31, 2022

END OF THE REPORT